Lecture 7

Channels: complete positivity and Stinespring

Objectives. *A part of a closed system need not evolve unitarily. Its state change must remain positive when the part is entangled with another system. In finite dimension, with trace preservation, that requirement is equivalent to a positive Choi matrix, a Kraus sum and a unitary evolution with an initially separate environment. We also construct the infinite-dimensional channels needed later.*

- Distinguish positivity from complete positivity using the transpose on half a singlet.
- Prove the finite Choi–Kraus–dilation equivalences, keeping tensor order and trace-preservation conditions explicit.
- Construct sufficient channels on a separable space and compute spontaneous-emission population and coherence changes.

7.1 Positivity must survive a spectator

A preparation with an ignored classical label gives a convex mixture. We assume a complex-linear evolution map on the span of density operators. In finite dimension that span is all matrices. In a separable $\mathcal{H}$, use L1's $\mathcal{T}_1(\mathcal{H})$, the complex span of positive trace-class operators. Complete positivity is a condition on that map, not on a single output state.

Definition 7.1 (Positive maps and channels)**.** A complex-linear map $\Phi : \mathcal{T}_1(\mathcal{H}) \to \mathcal{T}_1(\mathcal{H}')$ is *positive* if $\Phi(\rho) \geq 0$ for every positive trace-class ρ. For a finite spectator with basis $(|a\rangle)$, put $V_a\psi = \psi \otimes |a\rangle$ and write $Y_{ab} = V_a^\dagger Y V_b$. The *amplification* is

$$(\Phi \otimes \mathrm{id}_m)(Y) = \sum_{a,b=1}^{m} \Phi(Y_{ab}) \otimes |a\rangle\langle b| .$$

The map is *completely positive* (CP) if this amplification is positive for every finite m. A *channel* is CP and trace preserving (TP): $\operatorname{Tr}\Phi(X) = \operatorname{Tr} X$ for all $X \in \mathcal{T}_1$.

These blocks belong to $\mathcal{T}_1$: diagonal compressions of a positive trace-class operator are positive with finite trace, and off-diagonal blocks are complex combinations of compressions along $|j\rangle \pm |k\rangle$ and $|j\rangle \pm i\,|k\rangle$ by polarisation. General inputs follow from the positive-span definition. Reversing the tensor order gives the equivalent test $\mathrm{id}_m \otimes \Phi$: the swap of the two factors is unitary. Positivity together with trace preservation makes every density operator's output a density operator. CP further makes the local rule usable on arbitrary entangled preparations with a finite spectator, without negative probabilities.

Transposition $\tau(X) = X^{\mathsf{T}}$ in a fixed basis is positive: if $X \geq 0$, then $\langle z, X^{\mathsf{T}} z\rangle = \overline{\langle \bar{z}, X\bar{z}\rangle} \geq 0$. It is also trace preserving. Nevertheless take the singlet $\psi_- = (|01\rangle - |10\rangle)/\sqrt{2}$. In the ordered basis $00, 01, 10, 11$,

$$(\tau \otimes \mathrm{id}_2)(|\psi_-\rangle\langle\psi_-|) = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}.$$

The vector $(|00\rangle + |11\rangle)/\sqrt{2}$ has eigenvalue $-1/2$. Thus a positive map on isolated states need not be a physical local channel. A Wigner antiunitary on pure states does not provide a new complex-linear channel on one half of every entangled pair: its action on density operators is the transpose after a unitary change of basis, and the spectator test detects the obstruction.

Self-check 7.1

1. Which output event in the displayed partial transpose would have probability $-1/2$?
2. Does trace preservation repair its failure of positivity on the pair?

7.2 One finite matrix tests complete positivity

For this section $\mathcal{H} = \mathcal{H}' = \mathbb{C}^n$. Write $\mathsf{e}_{ij} = |i\rangle\langle j|$ and $\Omega = \sum_{i=1}^{n} |i\rangle \otimes |i\rangle$, which is *not normalised*: $\|\Omega\|^2 = n$. Define the *Choi matrix*

$$C_\Phi = \sum_{i,j} \mathsf{e}_{ij} \otimes \Phi(\mathsf{e}_{ij}) = (\mathrm{id}_n \otimes \Phi)(|\Omega\rangle\langle\Omega|). \tag{7.1}$$

The first factor is a reference copy of the input; the second is output. This order is bookkeeping for C_Φ. In a physical dilation below, the system is first and environment second, as elsewhere in the course.

Theorem 7.2 (Finite Choi–Kraus theorem)**.** *For a complex-linear map on $n \times n$ matrices the following are equivalent: complete positivity; $C_\Phi \geq 0$; and a finite* Kraus form

$$\Phi(X) = \sum_{k=1}^{d} K_k X K_k^\dagger, \qquad d \leq n^2.$$

Such a map is trace preserving exactly when $\sum_k K_k^\dagger K_k = \mathbb{1}$.

Proof. CP implies $C_\Phi \geq 0$ by its definition and positivity of $|\Omega\rangle\langle\Omega|$. If $C_\Phi \geq 0$, its finite spectral decomposition can be written $C_\Phi = \sum_{k=1}^{d} |v_k\rangle\langle v_k|$, absorbing square roots of the nonzero eigenvalues into v_k, so the v_k need not be unit vectors. For the zero matrix take a zero Kraus operator. Define matrices by reshaping:

$$(K_k)_{ai} = (v_k)_{(i,a)}.$$

The Choi entry with row (i,a) and column (j,b) is then

$$[\Phi(\mathsf{e}_{ij})]_{ab} = \sum_k (v_k)_{(i,a)}\overline{(v_k)_{(j,b)}} = \sum_k (K_k)_{ai}\overline{(K_k)_{bj}} = \left[\sum_k K_k \mathsf{e}_{ij} K_k^\dagger\right]_{ab}.$$

The matrix units form a basis, so the Kraus formula follows for every X. Conversely each amplified Kraus map sends a positive Y to $\sum_k (K_k \otimes \mathbb{1}_m) Y (K_k^\dagger \otimes \mathbb{1}_m)$, which is positive: each summand's quadratic form is nonnegative. Thus it is CP. Finally finite matrix cyclicity gives $\operatorname{Tr}\Phi(X) = \operatorname{Tr}(X \sum_k K_k^\dagger K_k)$. Equality to $\operatorname{Tr} X$ for every matrix unit forces the sum to be $\mathbb{1}$, and that sum conversely gives TP. ∎

For a channel, $\operatorname{Tr} C_\Phi = n$, so C_Φ/n is a density matrix. It is the output from half the normalised maximally entangled input $\Omega/\sqrt{n}$. Normalising it at the start without changing the reshaping formula would multiply every Kraus operator by $1/\sqrt{n}$ and lose trace preservation. Also $\sum K_k K_k^\dagger = \mathbb{1}$ is a different condition: it says the map sends $\mathbb{1}$ to $\mathbb{1}$, called *unitality*. Channels need not be unital.

An off-diagonal Kraus operator makes the row and column order visible. In the two-qubit order $00, 01, 10, 11$, take

$$v = |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = |1\rangle\langle 0| .$$

The first label of v becomes the column of K. The second becomes its row, as shown in Figure 7.1. The positive Choi matrix $|v\rangle\langle v|$ therefore gives the CP operation $X \mapsto KXK^\dagger$. Its completeness operator is $K^\dagger K = |0\rangle\langle 0|$, so this operation is not TP. It sends the input $|0\rangle\langle 0|$ to $|1\rangle\langle 1|$ and the input $|1\rangle\langle 1|$ to zero. A positive Choi matrix determines a CP map; trace preservation remains a separate test.

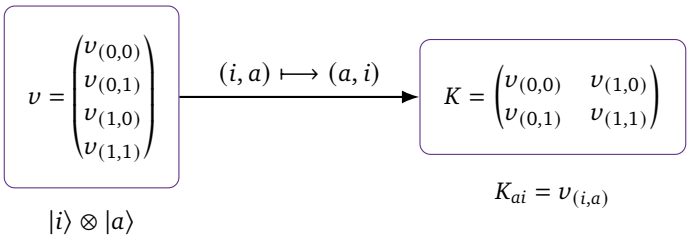


Figure 7.1: Reshaping one Choi vector in input-reference/output order. The input label i becomes a Kraus column and the output label a becomes a row; no component is complex conjugated.

For the identity channel, $C_{\mathrm{id}} = |\Omega\rangle\langle\Omega|$ has rank one; reshaping Ω gives the single Kraus operator $\mathbb{1}$. This provides a quick check of both tensor order and normalisation. Transposition's Choi matrix is the swap $\sum_{ij} e_{ij} \otimes e_{ji}$; its antisymmetric vectors have eigenvalue -1, so the theorem independently detects the failure found with the singlet.

Self-check 7.2

1. Reshape $v = |00\rangle + i|11\rangle$ into its Kraus operator. Is the resulting map TP?
2. Which completeness sum tests TP, and which tests unitality?

7.3 A channel is a unitary with an ignored environment

Theorem 7.3 (Finite Stinespring and unitary dilation). *A finite-dimensional map is CP exactly when $\Phi(X) = \mathrm{Tr}_E(JXJ^\dagger)$ for a linear $J : \mathbb{C}^n \to \mathbb{C}^n \otimes \mathbb{C}^d$. It is TP exactly when J is an isometry. Every finite-dimensional channel can be realised as*

$$\Phi(\rho) = \mathrm{Tr}_E\Big[U_E(\rho \otimes |e_\star\rangle\langle e_\star|)U_E^\dagger\Big] \tag{7.2}$$

with a fixed unit environment vector $e_\star$ and a unitary U_E. In the finite Kraus basis one may choose $e_\star = e_1$.

Here the partial trace is extended linearly from density operators to all finite matrices. On rank-one products it is $\mathrm{Tr}_E(|a \otimes b\rangle\langle c \otimes d|) = \langle d, b\rangle\, |a\rangle\langle c|$ (QM-I, Theorem 11.1); bounded local pairings make the extension well-defined.

Proof. For Kraus operators set $J\psi = \sum_k K_k\psi \otimes e_k$, with an orthonormal environment basis. The partial trace gives $\mathrm{Tr}_E(JXJ^\dagger) = \sum_k K_k X K_k^\dagger$ on rank-one matrix units, hence on all matrices. Conversely expand any linear J in environment coordinates to obtain those K_k. Moreover $J^\dagger J = \sum_k K_k^\dagger K_k$, proving the isometry equivalence. For a channel the rule $\psi \otimes e_\star \mapsto J\psi$ is an isometry between two n-dimensional subspaces of the same nd-dimensional space. Their orthogonal complements both have dimension $n(d-1)$. Map an orthonormal basis of one complement onto one of the other, obtaining a unitary U_E with the required action. This proves (7.2). Conversely that unitary rule gives an isometry J and hence a channel. ∎

The environment is prepared in one fixed state *independently* of ρ. Ignoring it loses information and can make the system mixed. Different dilations and Kraus lists can give the same channel; unitary freedom after padding is related to L2's freedom of ensembles (Theorem 2.5) (named: Nielsen–Chuang, Theorem 8.2, p. 372). No preferred list of physical unobserved events follows from the channel alone.

Initial correlations require the actual joint preparation. Compare $|\Phi_+\rangle\langle\Phi_+|$, $\Phi_+ = (|00\rangle + |11\rangle)/\sqrt{2}$, with $\mathbb{1}_4/4$. Both system and environment marginals are $\mathbb{1}_2/2$ in both preparations. Apply the fixed unitary $|s, e\rangle \mapsto |s \oplus e, e\rangle$, a controlled NOT with environment controlling the system. The Bell vector becomes $|0\rangle \otimes |+\rangle$, whereas $\mathbb{1}_4/4$ stays unchanged. The system outputs are $|0\rangle\langle 0|$ and $\mathbb{1}_2/2$. Only the correlation differed, yet neither marginal recorded it. There is no single output determined by the system marginal for this preparation class. One needs a specified joint-state assignment; this does not say every correlated case forbids every restricted-domain map.

In infinite dimension the orthogonal-complement argument needs a further hypothesis or a larger environment: two isomorphic infinite-dimensional subspaces need not have isomorphic complements. The theorem just proved is finite-dimensional. We next prove sufficient infinite constructions; the general representation theorem will remain cited.

Self-check 7.3

1. Construct J for $K_0 = \sqrt{1-p}\,\mathbb{1}$, $K_1 = \sqrt{p}\,\sigma_z$. Give $U_E\,|0, 1_E\rangle$ completing the two ready-state columns.
2. In the correlation example, which missing preparation information prevents a system-state map?

7.4 Sufficient constructions on a separable space

Let (K_k) be countably many bounded operators on a separable $\mathcal{H}$, with $\sum_k K_k^\dagger K_k = \mathbb{1}$ strongly. For positive trace-class $\rho = \sum_j p_j \left|\varphi_j\right\rangle\!\left\langle\varphi_j\right|$, the double rank-one family

$$\sum_{j,k} p_j \left|K_k\varphi_j\right\rangle\!\left\langle K_k\varphi_j\right| \quad \text{has total trace weight} \quad \sum_{j,k} p_j \left\|K_k\varphi_j\right\|^2 = \operatorname{Tr}\rho.$$

It defines a positive trace-class operator: its finite positive sums have operator-norm tails bounded by the omitted trace-weight tails; their norm limit is positive, and nonnegative diagonal sums give the stated trace. For every finite partial sum, its positive remainder has trace equal to the omitted weight, by the same nonnegative diagonal calculation. L1's bound $|\operatorname{Tr}(R_N A)| \le \|A\| \operatorname{Tr} R_N$ makes its bounded expectation tend to zero. Hence for bounded A the expectation is the series

$$\operatorname{Tr}(\Phi(\rho)A) = \sum_k \operatorname{Tr}(\rho K_k^\dagger A K_k). \tag{7.3}$$

The bound $\sum_k |\operatorname{Tr}(\rho K_k^\dagger A K_k)| \le \|A\| \operatorname{Tr}\rho$ follows termwise from the spectral expansion. This formula makes the output independent of the expansion and additive on positive inputs. Extend complex-linearly to L1's span $\mathcal{T}_1$. It is well-defined: any two decompositions have the same finite-sum pairings in (7.3), hence the same limits, and bounded expectations separate trace-class operators by their rank-one matrix elements.

Repeating the positive-series construction for $K_k \otimes \mathbb{1}_m$ proves CP: its (a,b) block is exactly $\sum_k K_k Y_{ab} K_k^\dagger = \Phi(Y_{ab})$, so it is the defined amplification. Their completeness follows first on finite spectator-coordinate vectors, then on all vectors, so the same total trace-weight identity proves TP. Thus the countable Kraus form defines a channel. In particular, $K_k = \sqrt{w_k}U_k$, with $\sum_k w_k = 1$ and unitary U_k, gives a discrete random-unitary channel.

For environment coordinates, the bounded map $\mathbb{1} \otimes \langle e_k|$ sends $\sum_\ell z_\ell \otimes e_\ell$ to z_k. An initially separate pure environment and an arbitrary unitary give $K_k\psi = (\mathbb{1} \otimes \langle e_k|)U_E(\psi \otimes e_\star)$. Parseval implies $\sum_k \|K_k\psi\|^2 = \|\psi\|^2$. The positive partial sums of $K_k^\dagger K_k$ are contractions; their quadratic tails tend to zero and therefore their strong tails do too, since $\|B\psi\|^2 \le \langle\psi, B\psi\rangle$ for $0 \le B \le \mathbb{1}$. This proves the required strong completeness. The partial trace in L6 gives the constructed channel: expand $U_E(\varphi_j \otimes e_\star) = \sum_k K_k\varphi_j \otimes e_k$; its bounded local expectation is $\sum_{j,k} p_j \left\langle K_k\varphi_j, AK_k\varphi_j\right\rangle$, which is (7.3). The general countable Kraus representation for the normal Heisenberg CP map associated with a trace-class channel is *cited*, not proved by a finite Choi matrix: Gustafson–Sigal, Theorem 19.21 p. 260 and Theorem 19.24 p. 261. An infinite unitary realisation may enlarge the ancilla to match complements.

The continuous construction below has a different input: a scalar average of expectations after unitary conjugation. First we check that this average is normal, so L1 constructs an output operator. Then the same construction with a finite spectator proves CP. Trace preservation follows from unitarity.

We will also need a continuous random kick. Here is a precise sufficient case, avoiding an unspecified operator integral. Suppose $\mathcal{W}(k)$ is a strongly continuous unitary family, $k \in \mathbb{R}$, and $w(k) \ge 0$ is a continuous probability density with improper integral one. For positive ρ define an expectation rule by scalar integrals:

$$\omega_\rho(A) = \int_{\mathbb{R}} w(k) \operatorname{Tr}(\rho \mathcal{W}(k)^\dagger A \mathcal{W}(k))\, dk. \tag{7.4}$$

The integrand without the density w is continuous and bounded by $\|A\| \operatorname{Tr}\rho$: finite spectral sums are continuous and their tails are uniformly bounded by $\|A\| \sum_{j>N} p_j$. Hence the

integral exists. Positivity is immediate and $\omega_\rho(\mathbb{1}) = \operatorname{Tr}\rho$. After normality is checked below, for $r = \operatorname{Tr}\rho > 0$ apply L1 to ω_ρ/r and multiply the resulting density operator by r; the zero input has zero output. We check normality explicitly.

For orthogonal projections with strong sum $E = \sum_j E_j$, put $B_N = E - \sum_{j\le N} E_j$. The continuous functions $h_N(k) = \operatorname{Tr}(\rho\mathcal{W}(k)^\dagger B_N \mathcal{W}(k))$ decrease to zero pointwise, by L1, and are nonnegative. They converge uniformly on every compact interval: for a chosen positive tolerance, continuity supplies a neighbourhood where one h_N is small at each point; finitely many such neighbourhoods cover the closed interval, and monotonicity gives one index working on the whole interval. Outside it the integral is bounded by $\operatorname{Tr}\rho$ times the probability-density tail. First choosing that tail small, then N large, proves $\int w h_N \to 0$. Thus (7.4) is normal and L1 constructs its positive trace-class output. Linearity and well-defined extension follow from the bounded pairings as above. Applying the same argument to finite spectators with $\mathcal{W}(k)\otimes\mathbb{1}_m$ proves CP: its blocks are exactly the integrated $\Phi(Y_{ab})$. Trace preservation follows from $\mathcal{W}(k)^\dagger\mathcal{W}(k) = \mathbb{1}$. This proves this continuous mixture is a channel; L11 uses a Gaussian density and $\mathcal{W}(k) = V(\hbar k) = e^{ikQ}$, whose strong continuity follows by a spatial-tail cutoff and uniform convergence on bounded positions.

Self-check 7.4

1. Verify strong completeness for $K_k = \sqrt{w_k}U_k$, with U_k unitary and $\sum_k w_k = 1$.
2. Why is existence of the scalar integral alone insufficient to invoke L1?

7.5 Spontaneous emission: population and coherence

Use QM-I's convention $H = (\hbar\nu_0/2)\sigma_z$, $\nu_0 > 0$: $|0\rangle = |\uparrow\rangle$ is excited and $|1\rangle$ is ground. An ideal zero-temperature emission channel with decay probability $0 \le \gamma \le 1$ has

$$K_0 = \sqrt{1-\gamma}\,|0\rangle\langle 0| + |1\rangle\langle 1|\,, \qquad K_1 = \sqrt{\gamma}\,|1\rangle\langle 0|\,.$$

Their completeness sum is $\mathbb{1}$, so they define a channel. We work in the frame rotating with H, omitting its free coherence phase; a thermal bath would also allow absorption. For $\rho = \begin{pmatrix} a & c \\ \bar c & 1-a \end{pmatrix}$,

$$\Phi_\gamma(\rho) = \begin{pmatrix} (1-\gamma)a & \sqrt{1-\gamma}\,c \\ \sqrt{1-\gamma}\,\bar c & 1-(1-\gamma)a \end{pmatrix}. \tag{7.5}$$

The excited population loses a factor $1-\gamma$; the coherence is multiplied by $\sqrt{1-\gamma}$. At $\gamma = 1$ every state becomes the ground state, so the channel is not invertible and is not a unitary conjugation. For $\gamma > 0$ it is not unital: the maximally mixed state moves toward the ground state.

A dilation makes the lost information explicit. With environment ready $|0_E\rangle$, take

$$J|0\rangle = \sqrt{1-\gamma}\,|0,0_E\rangle + \sqrt{\gamma}\,|1,1_E\rangle\,, \qquad J|1\rangle = |1,0_E\rangle\,.$$

The images are orthonormal, so this is an isometry; completing a finite basis supplies U_E. The two environment coordinates give exactly the displayed Kraus operators. For an initially excited system, the joint state stays pure; its system marginal becomes mixed for $0 < \gamma < 1$. Interpreting the environment labels as a detector's observed outcomes would add a reading procedure, studied in L10. Ignoring those labels already determines the channel.

If $\gamma(t) = 1 - e^{-t/T_1}$, the population decays as e^{-t/T_1} and coherence as $e^{-t/(2T_1)}$. Thus the pure-decay coherence time is $T_2 = 2T_1$. This exponential time dependence is a model

assumption, not a consequence of the fixed-time Kraus representation; L8 identifies the semigroup assumption and adds dephasing.

Self-check 7.5

1. Apply the channel with $\gamma = 3/4$ to $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.
2. Why does a pure joint output not force a pure system output?

What can happen to an open system?

A channel is a complex-linear, completely positive, trace-preserving state map. In finite dimension it is exactly a part of a unitary evolution with an independently prepared environment. Countable Kraus sums and the specified continuous unitary mixtures also define channels on separable spaces. An initially correlated preparation needs its own joint-state specification; a system marginal alone need not determine its future.

Summary. The spectator test excludes transpose as a local channel. A positive finite Choi matrix yields Kraus operators by explicit reshaping; their completeness sum makes the dilation an isometry and permits a finite unitary extension. Infinite sufficient constructions need convergence and normality, not a formal replacement of finite sums by integrals. Spontaneous emission reduces population and coherence by different factors while its system and environment together may remain pure.

Exercises. Try Sheet 7 in the separate *Exercise sheets* booklet (English).

Self-check answers.

7.1 1. The projection onto $(|00\rangle + |11\rangle)/\sqrt{2}$. 2. No; the negative expectation remains.

7.2 1. $K = \mathrm{diag}(1, i)$; yes, $K^\dagger K = \mathbb{1}$. 2. $\sum K_k^\dagger K_k = \mathbb{1}$ is TP; $\sum K_k K_k^\dagger = \mathbb{1}$ is unitality.

7.3 1. $J\psi = \sqrt{1-p}\psi \otimes |0_E\rangle + \sqrt{p}\sigma_z\psi \otimes |1_E\rangle$; a completing column is $-\sqrt{p}\,|0, 0_E\rangle + \sqrt{1-p}\,|0, 1_E\rangle$. 2. The joint correlation, which neither marginal determines.

7.4 1. The partial completeness sum is $(\sum_{k\le N} w_k)\mathbb{1}$, converging in norm to $\mathbb{1}$. 2. L1 represents normal expectation rules; countable projection additivity still needs proof.

7.5 1. $\begin{pmatrix} 1/8 & 1/4 \\ 1/4 & 7/8 \end{pmatrix}$. 2. Entanglement: the partial trace need not be a ray.

Sources. Watrous, *The Theory of Quantum Information*, Theorem 2.22 pp. 82–84 (opposite Choi factor order); Heinosaari–Ziman, *The Mathematical Language of Quantum Theory*, pp. 186–189, 207–209 (also an output-first convention in Theorem 4.48); Nielsen–Chuang, §8.2 pp. 360–376; Gustafson–Sigal, *Mathematical Concepts of Quantum Mechanics*, 3rd ed. (2020), Theorems 19.21, 19.24 pp. 260–265. The explicit infinite-series and continuous-expectation arguments above use L1 and L6.